

# Cultural Matching and the Persistence of Social Ties

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How do cultural tastes shape the evolutionary dynamics of personal networks during major life transitions? While traditional perspectives often view tastes primarily as malleable byproducts of network transmission, we advance an integrative cultural-matching framework in which pre-existing tastes and cultural practices actively guide tie retention. Using longitudinal ego-network data from the *Net-Sense* study, which tracked college students across eight survey waves, we find that shared cultural tastes—both broad categorical interests and specific shared activities—significantly reduce the hazard of tie dissolution. Conversely, cultural network opacity, or a lack of mutual cultural knowledge, acts as an independent liability that accelerates tie decay. Crucially, these stabilizing effects are heavily concentrated among subjectively close ties: cultural alignment forms an essential interactional core for maintaining core relationships amidst high background churn, whereas weak ties face such a high baseline rate of decay that cultural matching alone cannot sustain them. These findings demonstrate how, within high-turnover peer ecologies, individuals draw upon cultural dispositions as portable resources to selectively curate and stabilize their most meaningful social bonds.

# 1 Introduction

A central issue in the study of culture and networks concerns the co-evolution of social ties and cultural tastes (Lizardo, 2006; Edelmann and Vaisey, 2014). Sociological perspectives on taste diffusion often posit that an individual’s cultural profile is continuously molded by their current social network configuration (Mark, 1998b; McPherson, 2004). In this view, tastes diffuse across existing homophilous ties, which are themselves structured by socio-demographic similarity (Lazarsfeld and Merton, 1954; McPherson et al., 2001; Feld and Grofman, 2009). Under this network-transmission imagery, tastes are often construed as fluid, network-contingent contents liable to shift as local network configurations evolve (Mark, 2003). Yet longitudinal research reveals that personal networks are frequently in flux, with ties being formed, renegotiated, and dissolved—particularly during critical life-course transitions such as entry into college, job changes, or relocation (Burt, 2000; Wellman et al., 1997; Kleinbaum, 2018; Roy et al., 2022; Weiss et al., 2022; Kempnich et al., 2024; Vacchiano et al., 2024). This pervasive network turnover raises an important substantive question: if personal networks undergo substantial turbulence during formative transitions, what explains the relative durability of individuals’ cultural dispositions, and how do individuals actively deploy these cultural profiles to navigate volatile social ecologies (Bourdieu, 1984; Holt, 1998)?

In this paper, we evaluate the empirical implications of a *cultural matching model* of tie persistence within the dynamic setting of an undergraduate cohort. Rather than treating tastes solely as passive by-products of existing networks, we propose that pre-existing cultural dispositions serve as an active micro-interactional “social glue” that helps determine which network ties survive and which dissolve amidst high baseline churn (Burt, 2000). Under this framework, dyadic cultural matching (commonalities in tastes and leisure practices) increases the over-time stability and cognitive salience of social ties, operating in tandem with structural and dyadic factors such as subjective closeness, spatial propinquity, and tie duration. Using rich longitudinal ego-network panel data from the *NetSense* study over eight survey waves (Bahulkar et al., 2016, 2017; Hachen et al., 2022), we track the persistence of non-kin, on-campus peer ties across multiple college-year transitions. This empirical context provides a strategic window into relationship dynamics during emerging adulthood, capturing both the rapid reorganization of networks during the freshman transition and the subsequent consolidation of personal networks as graduation approaches.

Empirically, we examine the cultural matching mechanism along two interrelated dimensions. First, we establish the baseline protective effect of aggregate cultural matching—spanning both broad categorical domains and concrete leisure activities—on tie persistence. Second, we introduce the concept of “cultural network opacity”—the degree to which an ego lacks mutual cultural knowledge about an alter’s preferences (indicated by substantive “Don’t Know” reports)—as an independent predictor of tie decay. By evaluating these mechanisms in a discrete-time event history framework, we illuminate how cultural matching operates as an active selection and retention mechanism within dynamic peer environments.

The remainder of this article proceeds as follows. First, we outline the theoretical framework linking tastes, structural position, and tie persistence during life transitions. Next, we describe the NetSense data and our discrete-time event history modeling strategy. Finally, we present

the empirical results and discuss their theoretical implications for the study of homophily, network turnover, and cultural capital.

## 2 Theoretical Framework

### 2.1 Tastes and Social Networks: Two Perspectives

The relationship between cultural tastes and social networks has been conceptualized in two contrasting ways. The *network transmission perspective*, drawing on structural imagery (Carley, 1991), views tastes as malleable, network-contingent contents (Erickson, 1988, 1996; Mark, 1998a, 2003). In this view, social ties constitute the pre-existing “infrastructure” or “pipes” through which cultural practices diffuse (Borgatti, 2005; Erickson, 1996). Tastes are learned, adopted, and abandoned based on continuous exchange of cultural information and local bandwagon imitation within an actor’s immediate ego network (Mark, 2003). Because personal networks exhibit substantial over-time volatility—with between one and two-thirds of ties turning over annually in most personal networks (Morgan et al., 1997; Sutor et al., 1997)—the network transmission perspective implies that individual tastes must exhibit corresponding volatility.

In contrast, the *cultural capital perspective* conceptualizes tastes as highly durable, embodied dispositions acquired via early socialization (Bourdieu, 1984, 1990; Holt, 1998; Childress and Nault, 2019). Emphasizing the “interconvertibility” of different forms of capital (Bourdieu, 1986), this view sees cultural aptitudes not merely as by-products of ties, but as fundamental resources that help people form and sustain network connections. Here, networks are seen as fleeting structures built and maintained via interactions (DiMaggio, 1987). Consequently, individuals actively use pre-existing tastes to construct novel relationships or shed culturally incompatible ties, a dynamic formalized in the cultural matching model (Lizardo, 2006; Vaisey and Lizardo, 2010). In this framework, internalized cultural practices exert an independent causal effect on the formation of social networks because people self-select into ties based on the match between their tastes and those of others (Shalizi and Thomas, 2011).

Building on this insight, recent theoretical elaborations specify how matching and influence operate simultaneously as dual mechanisms. For instance, Lewis and Kaufman (2018), building on early work by Lizardo (2006) propose a generalized conversion model showing that cultural conversion (general connectivity as a result of having a wide variety of ties) and cultural matching (forming specific ties with others based on shared tastes) operate in tandem, constrained by the local cultural ecology (the distribution of particular tastes and cultural practices in the relevant setting). Crucially in our present study, the degree of cultural match at the dyadic level fundamentally determines whether relationships endure over time or instead selectively decay (Martin and Yeung, 2006). If tastes are durable and personal networks are volatile, it is highly likely that tastes serve as *drivers* of network connections rather than being determined exclusively by them. This leads to the concept of *choice homophily* or cultural matching: new social connections are selected and existing connections are maintained based on pre-existing cultural compatibility (Lazarsfeld and Merton, 1954; Werner and Parmelee, 1979). In other words, socio-demographic homophily may frequently operate as a *by-product* of active contact

selection keyed to cultural alignment, rather than as an exogenous structural constraint (Carley, 1991; DiMaggio, 1987).

## 2.2 Cultural Matching and Tie Persistence

A major challenge for any newly formed social relationship is what Burt (2000), drawing on Stinchcombe (1965), refers to as the “*liability of newness*.” Newly formed dyadic ties are fragile and exhibit a high baseline hazard of decay (“tie decay”), which steeply declines as a relationship matures and ripens into close friendship. What stabilizes a tie and helps it overcome this liability of newness? While classic network models emphasize structural factors (such as network constraint or multiplexity) and dyadic factors (such as subjective closeness), we propose that *cultural matching* is a vital stabilizing mechanism. Rather than merely indicating demographic similarity, shared tastes and cultural practices serve as an active element that helps people establish basic commonalities in outlook. At the micro-level, this matching stabilizes ties by reducing interactional friction, generating positive emotional energy during shared rituals (Collins, 2004), and providing a common cognitive and behavioral baseline that makes the tie more salient and rewarding to maintain (DiMaggio, 1987).

When ego and alter share cultural dispositions, they can engage in shared activities and coordinate behavior with minimal friction. Over time, this cultural alignment increases the cognitive salience of the alter, rendering the tie resistant to decay and facilitating its retention in the ego’s active network (Burt, 2000). Therefore, we expect that commonalities in cultural tastes and leisure practices will significantly reduce the hazard of tie decay, net of subjective closeness and structural controls (*The Baseline Matching Hypothesis*).

## 2.3 Cultural Network Opacity

If shared cultural knowledge stabilizes ties, then a *lack* of mutual cultural knowledge—*cultural network opacity*—should act as a major, independent mechanism of tie decay. We define cultural network opacity as the degree to which an ego is ignorant of an alter’s cultural preferences. It is important to differentiate opacity from mere “taste dissimilarity.” A taste mismatch implies that ego and alter know each other’s preferences but do not share them. Opacity, in contrast, implies a relational void. When an ego reports “Don’t Know” regarding an alter’s preferences across multiple domains, it signals a profound lack of interactional investment. Furthermore, it creates a severe cognitive barrier to finding focal points for shared activities. Without a map of the alter’s cultural landscape, the ego cannot easily initiate shared rituals or conversations, independently predicting a higher hazard of tie dissolution (Burt, 2002; Wellman et al., 1997). We test both the main effect of cultural network opacity on tie decay and the moderation of the protective effect of known cultural matches by opacity.

## 2.4 Tie Strength and the Limits of Cultural Matching

Furthermore, recent research highlights a critical theoretical tension regarding how the effects of cultural matching intersect with tie strength. On the one hand, drawing on the work of Edelman and Vaisey (2014) and Vaisey and Lizardo (2010), one might expect that cultural

matching is primarily associated with the maintenance of *strong* ties. In this view, already grounded friendships characterized by subjective closeness require ongoing cultural resonance and shared practices to sustain intimacy over time, making close ties highly sensitive to cultural matching (*The Strong-Tie Matching Hypothesis*).

On the other hand, a competing perspective, drawn from structural network theory, suggests that strong ties are highly resilient due to emotional inertia, multiplexity, and structural embeddedness, making them less dependent on specific cultural similarities for survival. Under this view, cultural matching acts as essential *scaffolding* specifically for *weak* or nascent ties. Weak ties suffer from a high baseline liability of newness (Burt, 2000); without shared interests to grease the wheels of interaction and provide focal points for engagement, they rapidly decay. Thus, the protective power of matching should be most pronounced for weak ties (*The Weak-Tie Scaffolding Hypothesis*). We estimate interaction models to adjudicate between these competing predictions.

## 2.5 Summary of Hypotheses

In sum, we test three core hypotheses regarding the role of culture in network evolution:

1. **Baseline Matching Hypothesis:** *Shared cultural tastes and activities significantly reduce the hazard of tie decay.*
2. **Cultural Network Opacity Hypothesis:** *A lack of mutual cultural knowledge independently increases the hazard of tie decay.*
3. **Moderation by Tie Strength:** We test two competing predictions regarding moderation:
  - **H3a (Strong-Tie Matching):** *The protective effect of cultural matching is amplified among subjectively close ties.*
  - **H3b (Weak-Tie Scaffolding):** *The protective effect of cultural matching is amplified among weak ties to overcome the liability of newness.*

## 3 Data and Methods

Data for this analysis comes from the *NetSense* study (Striegel et al., 2013), a longitudinal survey of college students conducted at the University of Notre Dame between 2011 and 2015. The study collected extensive panel data on students’ social networks, demographic background, and cultural preferences.<sup>1</sup> We collected information on the cultural habits of ego-alter pairs using both a series of closed-form domain questions and open-ended activity nominations. For the closed-form items, egos were asked to rate both their own and their alters’ level of interest across six broad cultural and leisure domains: *music*, *movies*, *books*, *sports*, *games*, and *outdoor activities*. For the open-ended items, respondents freely nominated up to five leisure activities

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<sup>1</sup>A comprehensive codebook, data dictionary, and repository for the NetSense project are available at <https://olizardo.github.io/NetSense-Codebook/>.

Table 1: Descriptive Statistics (Continuous Variables)

|                                         | Mean | SD   | Min  | Max   |
|-----------------------------------------|------|------|------|-------|
| Closed-Form Matches                     | 1.67 | 1.49 | 0.00 | 6.00  |
| Open-Ended Matches                      | 2.52 | 1.34 | 0.00 | 5.00  |
| Cultural Network Opacity                | 0.45 | 1.13 | 0.00 | 6.00  |
| Structural Embeddedness (Common Alters) | 6.52 | 3.84 | 0.00 | 19.00 |
| Tie Duration (Years)                    | 4.23 | 6.21 | 0.02 | 21.16 |

they enjoyed, and then indicated whether their alter also engaged in each nominated activity. Surveys were administered at the start and end of each semester via Qualtrics during the study period.

### 3.1 Study Waves

For this analysis of tie persistence, we pool data across adjacent waves to construct a person-period dataset. To isolate peer dynamics, we restrict our sample to reported non-kin, on-campus ties (i.e., alters who are also Notre Dame students). To capture relationship dynamics throughout the college experience, we construct a discrete-time person-period dataset spanning all available adjacent survey waves (Wave 1 through Wave 8). This design allows us to observe tie persistence continuously across varying intervals of the students’ academic careers, capturing both the initial network churn of freshman year and the consolidation of networks approaching graduation.

### 3.2 Measures and Descriptive Statistics

Tables 1 and 2 present summary statistics for all continuous and categorical variables used in our analyses across the  $N = 5,584$  dyad-period observations.

#### 3.2.1 Outcome Variable

*Tie Persistence*: A binary indicator taking a value of 1 if a social tie reported by an ego in a baseline wave (e.g., Wave 1) is still present in the subsequent wave (e.g., Wave 2), and 0 if the tie has dissolved (i.e., the alter is no longer listed in the ego’s network). In our analytic sample, 29.5 % of dyad-periods persist into the subsequent wave, reflecting the substantial baseline turnover characteristic of collegiate peer networks. To determine tie presence, egos were asked to list the people with whom they interact most frequently, using the following name generator: “Please list the people with whom you have spent the most time in the last month.” Egos could nominate up to 20 alters in each wave. This name generator uses what (Marin and Hampton, 2007) refers to as the “interaction” approach, which is relatively neutral with respect to tie strength (Laumann et al., 1989). Other tie characteristics related to strength, such as duration,

Table 2: Descriptive Statistics (Categorical Variables)

|                       |                | N    | Percent |
|-----------------------|----------------|------|---------|
| Tie Persisted         | 0              | 3935 | 70.47   |
|                       | 1              | 1649 | 29.53   |
| Ego Gender            | Men            | 2901 | 51.95   |
|                       | Women          | 2471 | 44.25   |
| Alter Gender          | Men            | 2839 | 50.84   |
|                       | Women          | 2728 | 48.85   |
| Subjective Closeness  | Not Close      | 565  | 10.12   |
|                       | Somewhat Close | 2611 | 46.76   |
|                       | Close          | 2400 | 42.98   |
| Interaction Frequency | Daily          | 4824 | 86.39   |
|                       | Weekly         | 760  | 13.61   |
| Race Homophily        | Heterophilous  | 510  | 9.13    |
|                       | Homophilous    | 1508 | 27.01   |
|                       | Unknown        | 3566 | 63.86   |

role relation (e.g., kin versus friend), and subjective closeness, were collected via follow-up name interpreters.

### 3.2.2 Main Predictors of Interest

*Closed-Form Cultural Matching:* A count variable (ranging from 0 to 6; Mean = 1.67, SD = 1.49) indicating the number of closed-form cultural domains in which the ego and alter share similar tastes. Egos were asked to rate their own interest, and their alters’ interest across six broad domains: music, movies, books, sports, games, and outdoor activities. The specific survey prompt for egos was: “How interested are you in the following activities?” and for alters: “How interested is this person in the following activities?” Response categories included “Very interested,” “Somewhat interested,” and “Not at all interested.” A domain was scored as a “match” if the ego rated both themselves and their alter identically. Note that the ego’s information was collected in a *separate* survey from the ego-network survey (a demographic survey), which dealt with immediate ordering bias and attenuated any cognitive consistency effects.

*Open-Ended Activity Matching:* A count variable (ranging from 0 to 5; Mean = 2.52, SD = 1.34) capturing the number of shared specific activity nominations. Egos were asked to freely list up to five leisure activities they enjoyed with the prompt: “Please list up to 5 specific activities that you enjoy doing in your free time.” Subsequently, for each alter, the ego was asked to check boxes indicating which of these specific activities the alter also enjoyed,

prompted with: “Which of your favorite activities does this person also enjoy?”

*Cultural network opacity*: A count variable (ranging from 0 to 6; Mean = 0.45, SD = 1.13) measuring the number of cultural domains for which the ego responded “Don’t know” regarding their alter’s preferences. For the closed-form domain questions regarding alters, respondents were provided a fourth response category: “Don’t know.” While survey researchers often instinctively treat “Don’t know” responses as missing data to be imputed or dropped, we explicitly operationalize these responses as substantive data. In the context of ego-network surveys, a “Don’t know” response is not random missingness; rather, it is a direct behavioral measure of relational ignorance (opacity). By retaining and counting these responses, we treat opacity as a distinct structural feature of the dyadic tie.

### 3.2.3 Additional Predictors

*Subjective Closeness*: Measured with the name interpreter, “How close are you to this person?” Response categories were “Not close” (10.1 %), “Somewhat close” (46.8 %), and “Very close” (43.0 %). This is treated as a factor variable in the models. As Marsden and Campbell (1984) noted in their classic paper, this item is the most reliable indicator of tie strength (Granovetter, 1973).

*Relationship Type*: Derived from the role relation name interpreter “How are you connected to this person?” We include binary indicators for whether the tie is a **Friend** (vs. other types) or a **Roommate/Dormmate** (capturing on-campus structural propinquity).

*Frequency of Interaction*: Measured with the name interpreter, “How often do you communicate with this person?” Responses were dichotomized into a **Daily** contact indicator (86.4 %) versus a reference category of **Weekly** contact (13.6 %).

*Tie Duration*: A continuous measure of how long the ego has known the alter (in years; Mean = 4.23 years, SD = 6.21), asked via the name interpreter: “How long have you known this person?” We include both the linear and squared terms (*Duration Squared*) to account for nonlinearities in the liability of newness.

*Structural Embeddedness (Common Neighbors / Triadic Closure)*: To account for the collective, group-embedded nature of personal networks and control for triadic closure (Granovetter, 1973; Coleman, 1988), we construct a dyadic embeddedness metric from the perceived alter-by-alter acquaintance matrix. In each survey wave, respondents evaluated whether their nominated contacts knew one another (“As far as you know, does [Alter Name] know any of your other contacts that are listed below?”). From this dyadic matrix, we compute the number of common neighbors (shared contacts) that Alter  $j$  possesses with other alters in ego’s network (Mean = 5.43, SD = 4.29; ranging from 0 to 19). Because the ego is tied to every alter in the roster, each reported alter-alter connection directly represents a closed triangle (Ego – Alter $_j$  – Alter $_k$ ). In our models, this measure is standardized. For wave intervals where the alter-alter matrix was not included (Wave 6), the variable is standardized using a non-parametric wave transition fixed effect to absorb the baseline shift, and sensitivity checks are conducted on the fully measured waves.

*Demographics*: We control for ego gender (44.3 % Women, 52.0 % Men) and alter gender (48.9 % Women, 50.8 % Men), along with a multiplicative interaction term to capture same-

gender vs. cross-gender dynamics. We also include a measure of *Race Homophily*, constructed by matching the ego’s self-reported race to the alter’s (27.0 % Homophilous, 9.1 % Heterophilous, 63.9 % Unknown alter race).

*Time Period*: Dummy variables for each wave transition interval (Wave 1 to 2, Wave 2 to 3, Wave 3 to 4, Wave 4 to 5, Wave 5 to 6, Wave 6 to 7, Wave 7 to 8) to allow the baseline hazard of tie persistence to vary non-parametrically across the entire panel span.

### 3.3 Analytical Strategy and Model Specification

Because our data tracks relationships over time and individuals can nominate multiple alters, our data structure consists of dyad-periods nested within egos. To properly analyze tie persistence, capture the high baseline volatility of personal networks (Cornwell et al., 2021; Weiss et al., 2022), and account for right-censoring, we employ a discrete-time survival analysis framework (event history modeling). Since the outcome variable—whether a tie persists into the next wave—is binary, we estimate the hazard of tie survival (the probability of persistence given the tie survived to the previous wave) using mixed-effects logistic regression. This modeling strategy allows us to explicitly model the baseline hazard rate of tie decay across each wave transition, isolating the protective effects of our cultural variables against this background volatility.

To address the non-independence of observations arising from the clustered nature of the ego-network data (multiple ties per ego), we include random intercepts for egos. This accounts for unobserved, time-invariant individual heterogeneity in the baseline propensity to maintain ties.

The models take the following general form:

$$\text{logit}(P(Y_{ijt} = 1)) = \beta_0 + \beta_1 X_{ijt} + \beta_2 C_{ijt} + u_i \quad (1)$$

where  $Y_{ijt}$  is the indicator of tie persistence for ego  $i$  and alter  $j$  at time  $t$ ,  $X_{ijt}$  represents the vector of our core theoretical predictors (cultural matching and cultural network opacity),  $C_{ijt}$  is a vector of dyadic and structural control variables (including time period indicators to allow the baseline hazard to vary non-parametrically over waves), and  $u_i$  is the ego-specific random intercept, assumed to be normally distributed with mean zero and variance estimated from the data.

All models are estimated via maximum likelihood using the `glmer` function from the `lme4` package in R. In models that interact subjective closeness with our cultural variables, we calculate and plot predicted probabilities to facilitate the interpretation of the non-linear interaction terms (Mize, 2019).

To verify the robustness of our core findings to modeling specification decisions, we conduct a supplementary analysis substituting our mixed-effects modeling framework for an ego fixed-effects framework using conditional logistic regression (Gaubard and Korn, 2011). This adjusts for all unobserved, time-invariant ego-level characteristics.

## 4 Results

We present our findings in three stages. First, we examine the baseline main effects of cultural matching and opacity on tie persistence. Second, we test the Strength-Mediated Matching Hypothesis by interacting our matching variables with subjective closeness. Finally, we assess the robustness of our findings using ego fixed-effects (conditional logit) models.

### 4.1 Main Effects: Cultural Matching, Cultural Network Opacity, and Structural Embeddedness

Table 3 presents the odds ratios from a series of mixed-effects logistic regression models predicting tie survival. As standard network theory would predict, ties that are subjectively closer ( $OR = 2.39, p < 0.001$  for “Close” ties in Model 1) are significantly more likely to survive. Interestingly, having daily contact ( $OR = 1.12, p > 0.05$ ) does not reach conventional significance levels when adjusting for closeness and other controls in this sample.

Building on this structural baseline, Model 1 shows that *closed-form cultural matching* significantly increases the odds of tie persistence. For each additional broad cultural domain in which an ego and alter share tastes, the tie is 9.6 % more likely to survive into the subsequent wave ( $OR = 1.10, p < 0.01$ ), net of controls.

In Model 2, we introduce *open-ended activity matches*. Specific, shared behavioral practices provide additive protective value: each shared activity increases the odds of tie survival by 8.0 % ( $OR = 1.08, p < 0.05$ ). When both are included, closed-form matching remains significant ( $OR = 1.08, p < 0.05$ ), showing that specific activities provide protective value above and beyond broad categorical taste similarities.

Model 3 tests the independent main effect of **cultural network opacity**—measured as the number of domains in which the ego reported “Don’t Know” regarding their alter’s preferences. Opacity predicts a 5.3 % decrease in the odds of tie persistence per unknown domain ( $OR = 0.95$ ). However, this main effect is not statistically significant at the conventional level ( $p > 0.05$ ) when adjusting for both closed and open-ended matches simultaneously. The attenuation of this coefficient in the pooled mixed-effects model is likely due to the built-in, between-person correlation between reporting “Don’t Know” and having fewer known closed-form matches. However, as we will show in the interaction models and the fixed-effects robustness check below, opacity remains a profound and significant liability within specific relational contexts.

Finally, Model 4 introduces **structural embeddedness**—the standardized count of common neighbors (shared alter contacts) that an alter possesses within ego’s nominated network. Consistent with triadic closure and collective embeddedness arguments, having shared network contacts strongly and significantly bolsters tie persistence ( $OR = 1.12, p < 0.05$ ): each standard deviation increase in common alter contacts increases the odds of tie survival by 12.2 %. Crucially, controlling for structural embeddedness does not eliminate the independent contribution of cultural matching. Both open-ended matches ( $OR = 1.07, p < 0.05$ ) and closed-form cultural matches ( $OR = 1.07, p = 0.057$ ) maintain their positive protective associations. This demonstrates that dyadic cultural affinity is not an artifact of being embedded in dense peer cliques; rather, culture and triadic structure operate as complementary, parallel anchors of

tie durability. Supplementary tests interacting cultural matching with structural embeddedness revealed no significant moderation ( $p = 0.65$ ), indicating that shared tastes protect ties across both structurally peripheral and densely embedded relationships.

To contextualize these coefficients, Figure 1 plots the average marginal effect of each core theoretical variable on the probability of tie persistence. Across the board, cultural matching serves as a protective mechanism against tie decay (positive estimates), whereas cultural network opacity acts as a liability (negative estimates).

## 4.2 The Moderating Role of Subjective Closeness

While the main effects show that cultural alignment generally stabilizes ties, we next assess whether this “social glue” is uniformly necessary for all relationships. We estimated auxiliary interaction models (not shown in the tables but visualized using predicted probabilities) to test whether the protective effects of cultural matching and the detrimental effects of cultural network opacity are moderated by baseline subjective closeness. Because interaction terms in logistic models can be opaque due to the non-linear link function, we visualize this critical interaction to make the core finding instantly digestible. Figure 2 plots the average marginal effects of our three cultural variables across ties rated as “Not Close,” “Somewhat Close,” and “Close.”

The results reveal a structured asymmetry in the protective effects of cultural matching and the deleterious effect of opacity conditioned on tie strength. We find a strong interaction for both closed-form cultural matches and cultural network opacity, but little interaction for open-ended matches. For broad, closed-form matches and network opacity, the predictive slopes (marginal effects) are steepest and most significant for *strong* ties (those rated “Close”). For these essential relationships, an increase in shared cultural tastes dramatically increases the predicted probability of tie persistence. Correspondingly, cultural network opacity has its most severe, statistically significant penalty precisely on these same strong ties: a lack of mutual cultural knowledge significantly accelerates their decay for ties that are rated as subjectively close. Conversely, for weak ties (those rated “Not Close”), the marginal effects for both closed-form matching and opacity are relatively flat, indicating an intrinsic fragility to decay regardless of cultural alignment or opacity in these broad domains.

This interaction suggests that shared cultural tastes and mutual cultural knowledge constitute the substantive “core” of deep, institutionalized friendships, requiring ongoing cultural alignment to be sustained. Weak ties, on the other hand, are fleeting and superficial; their baseline hazard of decay is so high that cultural matching does little to save them. In contrast, the protective effect of specific, open-ended activity matches operates more uniformly across relationships of all strengths, providing a consistent anchor for tie persistence, whether the tie is new or well-established.

## 4.3 Robustness Check: Ego Fixed-Effects

To confirm that our findings are not artifacts of our primary mixed-effects specification, we estimated an *Ego Fixed-Effects model* using conditional logistic regression (see Table

Table 3: Odds Ratios for Tie Persistence (Main Effects Models)

|                                            | Model 1<br>Closed   | Model 2<br>+ Open   | Model 3<br>+ Opacity | Model 4<br>+ Embed. |
|--------------------------------------------|---------------------|---------------------|----------------------|---------------------|
| Closed-Form Matches                        | 1.096**<br>(0.035)  | 1.083*<br>(0.035)   | 1.070*<br>(0.036)    | 1.067+<br>(0.036)   |
| Open-Ended Matches                         |                     | 1.080*<br>(0.034)   | 1.076*<br>(0.034)    | 1.071*<br>(0.033)   |
| Network Opacity                            |                     |                     | 0.947<br>(0.038)     | 0.952<br>(0.038)    |
| Structural Embeddedness<br>(Common Alters) |                     |                     |                      | 1.122*<br>(0.050)   |
| Subjective Closeness:<br>Somewhat          | 1.440**<br>(0.203)  | 1.388*<br>(0.197)   | 1.368*<br>(0.195)    | 1.342*<br>(0.191)   |
| Subjective Closeness: Close                | 2.388***<br>(0.359) | 2.244***<br>(0.342) | 2.188***<br>(0.336)  | 2.097***<br>(0.324) |
| Roommate/Dormmate                          | 0.788<br>(0.195)    | 0.787<br>(0.194)    | 0.783<br>(0.194)     | 0.791<br>(0.197)    |
| Friend                                     | 0.714<br>(0.175)    | 0.705<br>(0.173)    | 0.706<br>(0.174)     | 0.712<br>(0.177)    |
| Frequency: Daily (vs Weekly)               | 1.119<br>(0.146)    | 1.107<br>(0.145)    | 1.085<br>(0.143)     | 1.081<br>(0.143)    |
| Race Homophily: Same Race                  | 0.950<br>(0.122)    | 0.944<br>(0.122)    | 0.935<br>(0.121)     | 0.932<br>(0.120)    |
| Tie Duration (Scaled)                      | 0.796<br>(0.183)    | 0.790<br>(0.182)    | 0.768<br>(0.178)     | 0.740<br>(0.172)    |
| Tie Duration Sq (Scaled)                   | 1.238<br>(0.303)    | 1.262<br>(0.309)    | 1.288<br>(0.317)     | 1.339<br>(0.331)    |
| Num.Obs.                                   | 5349                | 5349                | 5349                 | 5349                |
| AIC                                        | 5147.9              | 5143.6              | 5143.7               | 5139.0              |
| BIC                                        | 5273.0              | 5275.3              | 5282.0               | 5283.8              |

+  $p \leq 0.1$ , \*  $p \leq 0.05$ , \*\*  $p \leq 0.01$ , \*\*\*  $p \leq 0.001$

Note: All models control for ego and alter gender, non-linear tie duration, and baseline hazard variations across all 7 wave transitions (coefficients omitted for space). Random intercepts for egos are included.

4). By stratifying the likelihood function by ego, this approach leverages only within-ego variance—comparing the alters a single ego retains with those it drops. This more stringent test more effectively isolates dyadic tie characteristics from unobserved individual baseline differences (e.g., an ego’s baseline sociability, introversion, or general propensity to use the “Don’t Know” survey option).

As shown in the table, the fixed-effects models yield results that are remarkably consistent with our random-effects main models. Closed-form matches ( $OR = 1.05, p < 0.10$ ) and open-ended matches ( $OR = 1.06, p < 0.05$ ) remain positive predictors of tie persistence, with odds ratios that are very close in magnitude to the baseline models. Crucially, in this strict within-ego specification, *cultural network opacity* emerges as a highly significant negative predictor of tie persistence ( $OR = 0.92, p < 0.01$ ), reducing the odds of tie survival by 8.2 % per unknown domain. This confirms that while opacity may correlate with overall match rates across individuals, *within* an individual’s personal network, a lack of cultural knowledge about a specific alter is a powerful, independent signal of tie dissolution. The structural controls—such as being “Close” ( $OR = 1.99, p < 0.001$ )—also maintain their consistent associations with the baseline hazard of tie decay.

## 5 Discussion and Conclusion

### 5.1 Summary of Key Findings

In this paper, we sought to address a central issue in the study of culture and networks: how do cultural tastes shape the evolutionary dynamics of personal networks? Using longitudinal ego-network data from the *NetSense* study, we considered the empirical implications of a cultural matching model of tie persistence. Our results provide strong evidence that shared cultural tastes are an important mechanism for stabilizing social ties over time, protecting them from natural decay processes (Burt, 2000). Specifically, both broad, closed-form cultural matches and specific, open-ended activity matches independently reduce the hazard of tie dissolution, even after rigorously controlling for dyadic structural embeddedness (triadic closure with other alters). While structural embeddedness itself strongly enhances tie survival ( $OR = 1.12, p < 0.05$ ), it does not displace cultural matching. Furthermore, we identified cultural network opacity—the lack of mutual cultural knowledge—as a yet-to-be-considered, but nevertheless independent relational liability that accelerates tie decay. Crucially, we found that the protective effect of closed-form matching and the detrimental effect of cultural network opacity are most pronounced among strong ties. For these subjectively close relationships, cultural alignment acts as a vital core for tie survival, whereas weak ties appear so fleeting that their baseline hazard of decay is high regardless of matching in these domains. In contrast, the protective effects of specific, open-ended matches remain relatively consistent across both weak and strong ties.

### 5.2 Theoretical and Substantive Interpretations

These findings carry significant theoretical implications for how we understand the relationship between tastes and networks, particularly regarding the role of tie strength. Our results heavily

Table 4: Robustness Models (Ego Fixed-Effects)

|                                         | Ego FE (clogit)     |
|-----------------------------------------|---------------------|
| Closed-Form Matches                     | 1.046+<br>(0.027)   |
| Open-Ended Matches                      | 1.056*<br>(0.025)   |
| Network Opacity                         | 0.920**<br>(0.029)  |
| Structural Embeddedness (Common Alters) | 1.055<br>(0.036)    |
| Subjective Closeness: Somewhat          | 1.409**<br>(0.155)  |
| Subjective Closeness: Close             | 1.949***<br>(0.233) |
| Roommate/Dormmate                       | 0.810<br>(0.149)    |
| Friend                                  | 0.806<br>(0.149)    |
| Frequency: Daily (vs Weekly)            | 1.078<br>(0.096)    |
| Race Homophily: Same Race               | 0.926<br>(0.081)    |
| Tie Duration (Scaled)                   | 0.820<br>(0.146)    |
| Tie Duration Sq (Scaled)                | 1.202<br>(0.220)    |
| Num.Obs.                                | 1640                |
| AIC                                     | 10669.8             |
| BIC                                     | 10777.8             |

+  $p \leq 0.1$ , \*  $p \leq 0.05$ , \*\*  $p \leq 0.01$ , \*\*\*  $p \leq 0.001$

Note: Controls for ego and alter gender, non-linear tie duration, and period fixed effects. Strata for egos are included.

favor the empirical implications of the *Strong-Tie Matching Hypothesis* over the *Weak-Tie Scaffolding Hypothesis*. Aligning with previous research (Edelmann and Vaisey, 2014; Vaisey and Lizardo, 2010), our longitudinal survival framework reveals that deep, institutionalized friendships require ongoing cultural resonance to sustain intimacy. It is the *strong* ties that are highly sensitive to cultural matching and opacity. Weak ties, suffering from a high liability of newness, rapidly decay with or without shared tastes. These results extend recent observations in life course and network research demonstrating that while significant network turnover is a ubiquitous feature of critical life transitions, individuals actively curate their core networks to maintain stable, supportive ties (Roy et al., 2022; Weiss et al., 2022; Kempnich et al., 2024; Vacchiano et al., 2024). More broadly, our findings challenge the assumption of the pure network-transmission perspective, which treats cultural tastes as merely fluid by-products of existing social ties (Mark, 2003). Instead, our findings align more closely with the cultural capital perspective (Bourdieu, 1984), suggesting that tastes function as durable, portable resources that individuals actively use to select, maintain, and discard social relationships.

Returning to the puzzle posed in the introduction—how do we account for durable cultural profiles in the face of volatile networks (Bourdieu, 1984; Holt, 1998)—our results help to close the loop. If persons use relatively stable cultural dispositions to selectively maintain their networks, it suggests that cultural capital is not merely a passive status marker or a “thin” effluvium of network position. Rather, it is an active navigational tool for managing social environments. People churn through weak ties at a high baseline rate, but they rely on their durable cultural compass to actively curate and protect the core relationships that matter.

Furthermore, our findings speak directly to the classic distinction between choice homophily and induced homophily (McPherson et al., 2001). If cultural matching actively prevents tie decay and opacity accelerates it, then a substantial portion of the observed homophily in established social networks is likely due to selective retention (choice homophily) rather than pure structural constraint or network-induced adaptation (Lizardo, 2006; Lewis and Kaufman, 2018). By retaining culturally matched ties and shedding incompatible ones over time, networks become homophilous through an active evolutionary sorting process.

In this paper, we conceptualize cultural matching as the essential structural core of close relationships. While early network theories emphasized structural constraints or multiplexity (Burt, 2000), our findings suggest that shared tastes may be what gives close relationships their staying power. In the absence of mutual cultural knowledge (high opacity), even subjectively close ties face a higher hazard of dissolution. This developmental view of networks helps reconcile the volatility of personal networks with the relative stability of individual cultural profiles.

Beyond network theory, these findings carry practical implications for institutions aiming to foster social cohesion. For university administrators and community organizers seeking to reduce social isolation, our results underscore that mere proximity or shared structural affiliations are insufficient to sustain durable relationships. Facilitating environments where individuals can discover and activate shared cultural interests—from niche clubs to diverse recreational programs—may be essential for transforming fleeting initial encounters into resilient, long-term support networks.

### 5.3 Limitations and Future Research

Despite these promising findings, several limitations point toward fruitful avenues for future research. First, our empirical focus on a collegiate cohort during emerging adulthood delineates specific boundary conditions for our claims. The college transition represents a uniquely dynamic developmental and institutional ecology characterized by high baseline network churn, rapid relational renegotiation, and geographic concentration. While this setting provides an ideal testing ground for observing tie decay and selective retention in real time, future research should examine whether cultural matching operates similarly as a stabilizing force across different life stages and institutional domains. In particular, comparative studies in workplace organizations, civic associations, or among older adults (Cornwell et al., 2008) could illuminate how the protective capacity of cultural matching varies across differing levels of structural constraint.

Second, an important question concerns the *temporal horizon* of cultural matching beyond the collegiate context. Our panel follows students through their undergraduate trajectory, capturing the critical period when campus ties are actively formed and pruned. However, a compelling open question is whether cultural matching continues to sustain social ties across major post-collegiate life transitions—such as graduation, geographic dispersal, labor market entry, and family formation. When individuals no longer share the daily scaffolding of an institutional campus environment, does deep cultural alignment provide the necessary conversational and ritual currency to keep distant ties alive across decades? Alternatively, do enduring collegiate ties persist primarily through accumulated relational history and institutional memory, rendering cultural matching less decisive once physical co-presence ends? Future long-term multi-decade panel studies will be vital for determining the life-course boundaries of cultural matching.

Third, our network measures capture the *ego-perceived* relational landscape. While egocentric network data are uniquely suited for measuring an actor’s cognitive orientation, subjective closeness, and perceived cultural knowledge about their alters, they capture only one side of the dyad. In non-reciprocated or unidirectional ties, relational orientations may be asymmetric, reflecting underlying differences in social status, popularity, or aspirational connection. In such cases, an ego may actively work to maintain a culturally matched tie with a higher-status peer even if the alter does not reciprocate the nomination. Although our ego fixed-effects and dyadic controls account for an ego’s general nomination patterns, future sociocentric investigations with complete dyadic reciprocity data can formally evaluate how cultural matching interacts with status differentials and reciprocal confirmation.

Fourth, the current operationalization aggregates cultural matches across broad domains of lifestyle and taste. Future research could further disaggregate these domains to examine whether high-status cultural genres (e.g., classical music, highbrow literature) confer distinct relational advantages compared to widely popular or subcultural domains (DiMaggio, 1987). In addition, while our measures focus primarily on positive cultural alignment and lack of knowledge (opacity), future work should systematically investigate the role of shared cultural *dislikes* and mutual distastes as distinct symbolic boundaries in tie dissolution.

Finally, while our discrete-time event history framework provides a rigorous foundation for modeling the hazard of tie decay net of unobserved heterogeneity, future investigations

with continuous digital trace streams would benefit from estimating stochastic actor-oriented models (SAOMs) or continuous-time network dynamics ([Hachen et al., 2022](#)). Such models can simultaneously estimate the reciprocal, co-evolutionary processes through which social ties induce taste convergence (influence) and pre-existing cultural dispositions guide tie selection and pruning (selection).

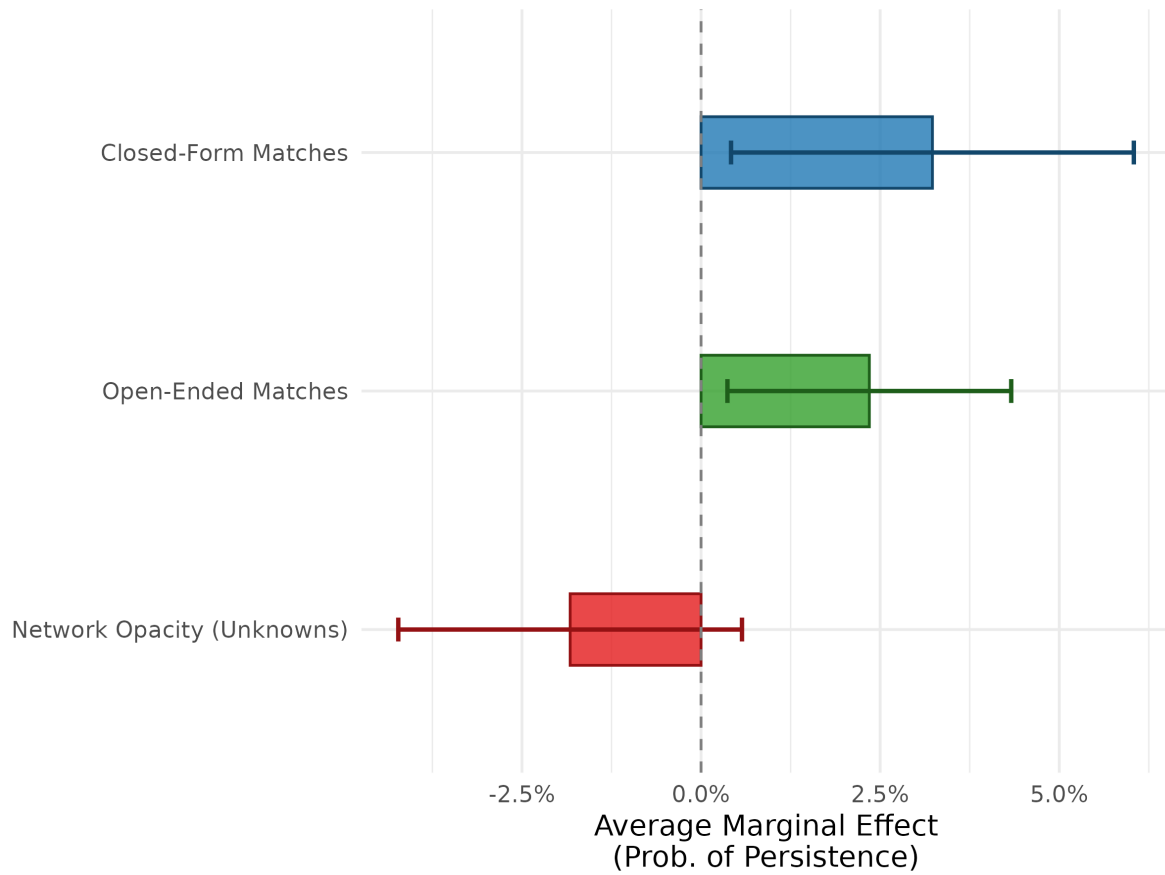


Figure 1: Predicted Probability of Tie Persistence by Cultural Matching and Cultural Network Opacity.

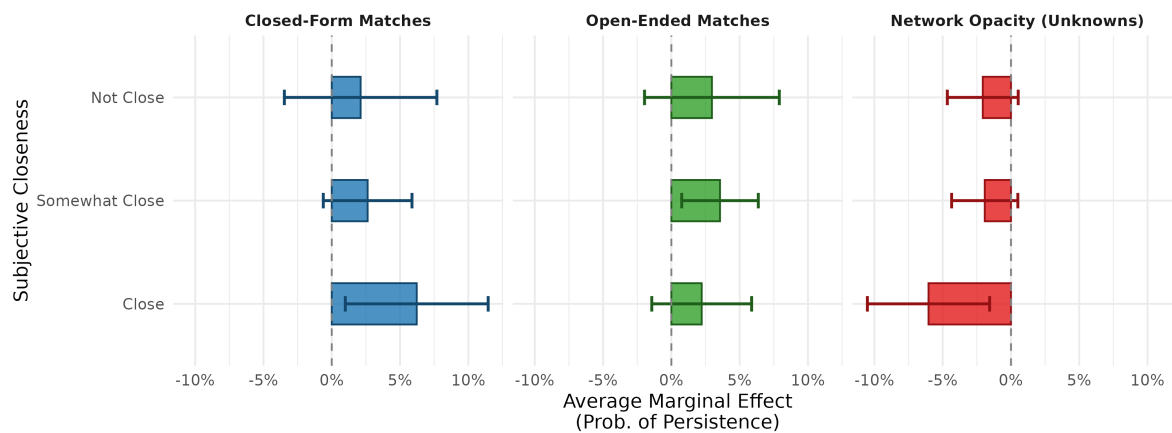


Figure 2: Predicted Probability of Tie Persistence by Cultural Matching, Cultural Network Opacity, and Subjective Closeness.

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